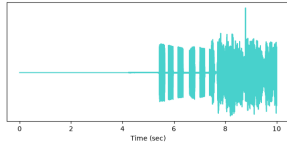


Example

Event Prediction

Filename: Wsa0JT2SsPg.flac



PretrainedSED

Detection			
Event	Onset	Offset	Confidence
Background noise	3.92	10.00	0.25
Mechanisms	4.08	10.00	0.13
Music	4.24	10.00	0.34
Telephone dialing, DTMF	5.36	8.32	0.10
Male singing	7.40	10.00	0.45

Prompts with style - {Flat List, Temporal Narrative, Scene-Structured, Chain-of-Thought (CoT)} (Common for all models)

Flat List

Model Input

Audio clip duration: 10.0 seconds.

Detected sound events:

- Background noise (3.92s – 10.00s, confidence: 0.25)
- Mechanisms (4.08s – 10.00s, confidence: 0.13)
- Music (4.24s – 10.00s, confidence: 0.34)
- Telephone dialing, DTMF (5.36s – 8.32s, confidence: 0.10)
- Male singing (7.40s – 10.00s, confidence: 0.45)

Write a single paragraph describing this audio scene in natural English.

Temporal Narrative

Model Input

Audio clip duration: 10.0 seconds.

Detected sound events (in chronological order):

- Background noise (3.92s – 10.00s, confidence: 0.25)
- Mechanisms (4.08s – 10.00s, confidence: 0.13)
- Music (4.24s – 10.00s, confidence: 0.34)
- Telephone dialing, DTMF (5.36s – 8.32s, confidence: 0.10)
- Male singing (7.40s – 10.00s, confidence: 0.45)

Write a single paragraph that describes the audio scene chronologically. Start from the beginning of the clip and narrate events in the order they occur. Use temporal connectives such as 'at first', 'then', 'followed by', 'meanwhile', 'at X seconds', 'by the end'. Do not skip any listed event.

Scene Structured

Model Input

Audio clip duration: 10.0 seconds.

All detected events:

- Background noise (3.92s – 10.00s, confidence: 0.25)
- Mechanisms (4.08s – 10.00s, confidence: 0.13)
- Music (4.24s – 10.00s, confidence: 0.34)
- Telephone dialing, DTMF (5.36s – 8.32s, confidence: 0.10)
- Male singing (7.40s – 10.00s, confidence: 0.45)

Background sounds (sustained, long-duration):

- Background noise (3.92s – 10.00s, confidence: 0.25)
- Mechanisms (4.08s – 10.00s, confidence: 0.13)
- Music (4.24s – 10.00s, confidence: 0.34)

Foreground events (short, prominent):

- Telephone dialing, DTMF (5.36s – 8.32s, confidence: 0.10)
- Male singing (7.40s – 10.00s, confidence: 0.45)

Write a single paragraph describing this audio scene. Begin with the background soundscape, then describe the foreground events as they occur in time. Only describe sounds from the lists above.

Chain-of-Thought (CoT)

Model Input

Audio clip duration: 10.0 seconds.

Detected sound events:

- Background noise (3.92s – 10.00s, confidence: 0.25)
- Mechanisms (4.08s – 10.00s, confidence: 0.13)
- Music (4.24s – 10.00s, confidence: 0.34)
- Telephone dialing, DTMF (5.36s – 8.32s, confidence: 0.10)
- Male singing (7.40s – 10.00s, confidence: 0.45)

Step 1 — Count and categorize: How many distinct events are there? Which overlap in time?

Step 2 — Identify the dominant sound and any transient events.

Step 3 — Write a single-paragraph natural language description of the scene that accurately reflects the temporal structure above.

Provide Step 1, Step 2 briefly, then Step 3 as the final paragraph.

Model: SmoLM2-135M

Flat List

Model Output

This audio scene features three distinct components: a background noise, mechanisms, and music. The background noise includes a soft hum from a nearby speaker, while mechanisms include various types of machinery such as motors, gears, and valves.

Temporal Narrative

Model Output

The audio clip begins at the moment when background noise ceases, followed by mechanisms that signal the start of the music. The mechanism then shifts to the subject of male singing, which continues for approximately two minutes before the final segment of the song ends. At this point, the main action takes place, with the speaker's voice rising dramatically as he performs the melody. This sequence unfolds over several seconds, punctuated by the occasional pause or silence caused by the telephone dialing process.

Scene Structured

Model Output

The scene unfolds in a serene and intimate setting, where the soft hum of conversation punctuates the quiet moments of nature's beauty. The background noise, a steady stream of background music, serves as a soothing background melody that calms the protagonist's mind and creates a sense of tranquility. The mechanisms, which include the gentle tapping of keys on the keyboard and the rhythmic cadence of the piano, add depth and texture to the overall atmosphere.

Chain-of-Thought (CoT)

Model Output

The scene consists of three distinct events: a background noise, mechanisms, and music. The background noise includes 3.92 seconds of background noise, while the mechanisms include 4.08 seconds of mechanical machinery and 5.36 seconds of musical notes. The music comprises 4.24 seconds of instrumental vocals and 7.40 seconds of vocal harmonies. The male singing segment has 7.40 seconds of vocal harmony and 10.00 seconds of melodic accompaniment.

To accurately capture the temporal structure, I will count and categorize each event separately. The background noise includes 3.92 seconds of background noise, while the mechanisms include 4.08 seconds of mechanical machinery and 5.36 seconds of musical notes. The music comprises 4.24 seconds of instrumental vocals and 7.40 seconds of vocal harmonies. The male singing segment has 7.40 seconds of vocal harmony and 10.00 seconds of melodic accompaniment.

I have identified three distinct events: a background noise, mechanisms, and music; two events involving mechanisms and one between mechanisms and music; and a third event consisting of a man singing. These categories provide a clear understanding of the sequence of events within the scene.

Model: SmoLM2-360M

Flat List

Model Output

The audio clip begins with background noise and then shifts to mechanisms, followed by music, which gradually transitions to male singing. The silence before the start of the musical section is notable for its high confidence level.

Temporal Narrative

Model Output

At first, there was background noise, which gradually increased over the next few seconds. Then, mechanisms started playing, followed by music. The telephone dialing, DTMF, and male singing were also present at some point during this period. Meanwhile, the scene continued to unfold, with more background noise emerging later on. By the end of the clip, all these elements had been incorporated into the overall narrative, creating a cohesive and immersive experience for the listener.

Scene Structured

Model Output

The background noise is sustained for approximately three and a half minutes, punctuated by brief moments of steady, low-level humming. The mechanisms event begins around two minutes later, followed by a series of short, distinct tones that gradually become more pronounced over the next few seconds. As the music picks up, it becomes clear that the main focus of the scene shifts towards a male singer who sings a catchy melody within a minute or so.

Chain-of-Thought (CoT)

Model Output

Step 1: Count and categorize

There are two distinct events: "Background noise" and "Mechanisms." The timing between these events is approximately 10.0 seconds.

Step 2: Identify the dominant sound and any transient events

The dominant sound is "Background noise," which has a higher confidence score at 0.25 out of 1. This means it is more likely to be present than other events. However, there is also a transient event, "Music," which has a lower confidence score at 0.13 out of 1.

Step 3: Write a single-paragraph natural language description of the scene

The scene consists of a background noise followed by mechanisms, music, and finally, a telephone dialing, DTMF, male singing, and then a silence before the last event ends. The overall temporal structure is roughly linear, with each event building upon the previous one.

Model: SmoLM2-1.7B

Flat List

Model Output

The audio scene begins with background noise at approximately 3.92 seconds, lasting for about three quarters of a second before it fades away.

Temporal Narrative

Model Output

At first, there was background noise present throughout the entire duration of the clip, which lasted for 10 seconds. Then, mechanisms were heard at around 4 seconds, followed by music at approximately 4.24 seconds. Meanwhile, telephone dialing, DTMF, could be heard starting at 5.36 seconds and continuing until 8.32 seconds.

Scene Structured

Model Output

The audio scene begins with a steady background noise that persists throughout the 10-second duration, providing a constant ambiance. The music starts around 4.24 seconds and continues until the end, adding a melodic element to the overall atmosphere. As the male singer takes over at 7.40 seconds, his voice becomes more prominent, creating a clear distinction between the different elements of the scene.

Chain-of-Thought (CoT)

Model Output

Step 1 — Count and categorize: There are six distinct sound events: background noise, mechanisms, music, telephone dialing, DTMF, and male singing. The overlapping events are background noise and mechanisms (start at 3.92s), music and telephone dialing (start at 4.08s), and DTMF and male singing (start at 7.40s).

Step 2 — Identify the dominant sound and any transient events. The dominant sound is male singing, which starts at 7.40s and lasts for approximately 2.68 seconds. Transient events include the start of each sound event and their transitions.

Step 3 — Write a single-paragraph natural language description of the scene that accurately reflects the temporal structure above. Here's the description: "A man sings softly in the background while various mechanical noises can be heard throughout the scene. A brief period of silence follows before a telephone begins to ring, followed by the ringing tone and subsequent DTMF digits being pressed."

Model: Qwen2.5-0.5B

Flat List

Model Output

The audio clip lasts for approximately 10 seconds and features various sounds including background noise, mechanisms, music, telephone dialing, male singing, and more.

Temporal Narrative

Model Output

At first, there's background noise that starts at 3.92 seconds and ends at 10.00 seconds. Then, we have mechanisms that start around 4.08 seconds but end around 10.00 seconds. Following this, music begins at 4.24 seconds and lasts until the end.

Scene Structured

Model Output

The audio clip begins with a series of background noises and mechanisms, lasting for approximately 3.92 seconds. The music starts around 4.08 seconds, followed by telephone dialing, DTMF signals, and male singing, which lasts for about 4.24 seconds each. The background sounds continue for another 5.36 seconds before transitioning to foreground events such as telephone dialing, DTMF signals, and male singing, which last for approximately 8.

Chain-of-Thought (CoT)

Model Output

Step 1: There are five distinct sound events in this audio clip:

- Background noise:** lasting from 3.92 seconds to 10.00 seconds.
- Mechanisms:** lasting from 4.08 seconds to 10.00 seconds.
- Music:** lasting from 4.24 seconds to 10.00 seconds.
- Telephone dialing, DTMF:** lasting from 5.36 seconds to 8.32 seconds.
- Male singing:** lasting from 7.40 seconds to 10.00 seconds.

Step 2: The predominant sound is the male singing, which lasts for approximately 7.40 seconds to 10.00 seconds.

Step 3: A single-paragraph natural language description of the scene that accurately reflects the temporal structure above would be:

"The male singer's voice fills the air as he sings on the phone, blending seamlessly between the background noise, mechanisms, music, telephone dialing, and finally, the male singing."

Model: Qwen2.5-1.5B

Flat List

Model Output

The audio begins with background noise, which continues throughout the entire recording period.

Temporal Narrative

Model Output

The audio begins with background noise, which persists throughout the entire recording period. At approximately three and a half seconds, there's a faint hum or buzz that gradually increases in volume until it reaches its peak at four and a quarter seconds. This is followed immediately by mechanisms operating quietly, indicating some kind of mechanical activity or equipment running smoothly. By five and thirty-six seconds, we hear telephone dialing, specifically DTMF tones, suggesting someone is making a phone call using touch-tone buttons.

Scene Structured

Model Output

The audio begins with a steady hum of background noises, punctuated by occasional mechanisms and music playing softly in the background. The atmosphere gradually shifts around 4.08 seconds when telephone dialing, DTMF tones begin to ring out, creating a distinct buzzing sound. This is followed closely by male vocals at 7.40 seconds, which quickly become more prominent until it fades away at 10.00 seconds. The overall tone remains calm but engaged throughout the recording period.

Chain-of-Thought (CoT)

Model Output

Step 1: There are four distinct sound events recorded:

- Background Noise:** Occurs from 3.92 seconds to 10.00 seconds.
- Mechanisms:** Also occurs from 3.92 seconds to 10.00 seconds.
- Music:** Starts at 4.24 seconds and continues until 10.00 seconds.
- Telephone Dialing, DTMF:** Begins at 5.36 seconds and ends at 8.32 seconds.

These events do not overlap significantly; they occur sequentially without interruption.

Step 2: The dominant sound appears to be music, which starts around 4.24 seconds and lasts through the entire recording period. Transient events include telephone dialing, which begins approximately halfway through the recording and concludes just before the end.

Step 3: A single-paragraph natural language description of the scene would be:

The recording captures a continuous background environment punctuated by intermittent musical compositions. Initially, the ambient noise is predominant, followed by a series of mechanisms operating within the space. Around 4.24 seconds, a clear melody begins, lasting throughout the entirety of the recording. This melody gradually fades out towards the end, leaving behind a final note on the telephone dialing tones, which conclude the sequence of events.